

Full Deep PRD / Product Blueprint / Team Execution Reference

Hybrid document in Arabic + English

Prepared from the accumulated project context, decisions, assumptions, and execution planning

Purpose: team alignment, client-facing structure, implementation guidance, and AI handoff baseline

Project focus	AI-powered high-school physics learning platform for Oman
Current scope	Web platform MVP with bilingual-ready UX, exams, dashboards, AI-assisted workflows
Primary users	Students, Teachers, Admins
AI strategy	Gemini-first, provider-flexible, backend-controlled
Database direction	PostgreSQL with question bank + attempts + evaluation model
Delivery use	Reference for team execution, client understanding, future SRS / costing

Status: working strategic document, not yet a legally signed final requirements baseline.

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Temporary Name	Target Level	Deployment Philosophy
المسرات / Al-Masarrat	High School for now	MVP first, future-ready architecture

1. Executive Summary | الملخص التنفيذي

Al-Masartrat is an AI-powered educational platform focused initially on physics for high school students in Oman. The platform combines exams, feedback, teacher productivity tools, structured progress tracking, and controlled AI assistance.

منصة المسرّرات هي منصة تعليمية ذكية تركز في مرحلتها الحالية على الفيزياء لطلاب المرحلة الثانوية في سلطنة عمان. الفكرة ليست مجرد نظام امتحانات، بل بيئة تعليمية تساعد الطالب على الفهم، وتساعد المدرس على الإنشاء والتقييم والمتابعة بكفاءة.

This document consolidates the working project vision, confirmed assumptions, recommended architecture, database direction, AI strategy, user flows, team system, delivery model, and practical MVP boundaries.

2. Project Identity and Scope | هوية المشروع والنطاق

Project Identity

- Project Name (temporary): منصة المسرّرات / Al-Masartrat
- Product Type: AI-assisted educational web platform
- Current Subject Focus: Physics
- Target Geography: Oman
- Primary Delivery Form: Web platform first, mobile later if validated

Current Scope

The current project direction is to build a professional MVP that proves the educational and operational value of the platform before expanding into a larger ecosystem.

- Bilingual-ready user experience (Arabic and English)
- Student dashboard, teacher dashboard, admin access
- Exam creation, question bank, attempts, results, comments or notes layer
- AI-assisted question generation and AI-assisted evaluation for suitable answer types
- Future-friendly technical structure rather than one-off prototype decisions

3. Problem, Vision, and Value Proposition | المشكلة والرؤية والقيمة

Problem Statement

Students often receive grades without understanding. Teachers often spend large amounts of time preparing questions, correcting answers, and monitoring many learners individually. Traditional exam systems usually stop at score display instead of educational feedback.

الطالب قد يحفظ أو يحل دون فهم واضح لمواضع الخطأ. والمدرس يستهلك وقتًا كبيرًا في بناء الأسئلة والتصحيح والمتابعة الفردية. كما أن كثيرًا من الأنظمة التعليمية تكتفي بعرض النتيجة دون تفسير أو تغذية راجعة مفيدة.

Vision

Build a practical and scalable AI-assisted learning platform that improves student understanding, increases teacher efficiency, and creates a disciplined educational workflow that can grow over time.

Value Proposition

For Students	For Teachers	For the Platform / Business
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Practice, feedback, clarity, progress visibility	Faster question creation, lower correction burden, clearer monitoring	Differentiated AI-assisted product with scalable operational logic
Explanations instead of raw scores only	Ability to review and override AI outputs	Foundation for future monetization and expansion

4. User Segments and Personas | المستخدمين والشخصيات الأساسية

Primary User Segments

Segment	Core Need	Typical Actions	Priority in MVP
Students	Practice, feedback, exam access	Take exams, review results, track progress	High
Teachers	Creation, evaluation, monitoring	Generate questions, build exams, review attempts	High
Admins	Control, access, oversight	Manage users, roles, activity and health of system	Medium

Persona Notes

- Student persona: needs clarity, fairness, simplicity, and visible improvement over time.
- Teacher persona: needs speed, control, academic confidence, and reduced repetitive work.
- Admin persona: needs role control, user visibility, and predictable operational behavior.

5. Product Scope: MVP vs Full Platform | نطاق المنتج: النسخة الأولى مقابل المنصة الكاملة

MVP - Build Now	Full Platform - Later Expansion
Authentication and role-based access	Mobile application
Student / teacher / admin core dashboards	Payment and subscription flows
Question bank foundation	Advanced analytics and reporting
Exam creation and publishing	Adaptive learning engine
MCQ grading + short-answer AI evaluation	Multi-subject expansion
AI question generation drafts for teachers	Richer notifications and communication system
Results, explanations, and progress basics	More advanced AI tutoring features
Bilingual-ready UX foundation	

6. Functional Requirements | المتطلبات الوظيفية

Student Module

- Register / sign in securely and reach the correct role-based area
- View available exams, previous attempts, scores, and progress indicators

- Open an exam, answer questions, submit the attempt, and view results
- Receive explanation, feedback, and improvement hints where appropriate

Teacher Module

- Create questions manually with type, difficulty, topic, hint, and explanation
- Request AI-generated question drafts with topic, difficulty, language, count, and type
- Review, edit, approve, reject, and publish generated content
- Create and manage exams from manual and AI-assisted question pools
- Track student performance and override AI outputs when necessary

Admin Module

- Manage users and account activity
- Control roles and visibility
- Review operational health and later analytics dashboards

Acceptance Logic for Core Features

Feature	Done when...	Notes
Login	Valid sign-in, invalid credential handling, role-based redirect	Secure session required
Exam submission	Answers stored, attempt status updates, grading flow triggered	MCQ direct + short-answer AI where enabled
AI generation	Teacher request accepted, valid JSON returned or safely rejected, drafts reviewable	Teacher remains final authority
Result view	Score, correctness level, and explanation visible cleanly	Student-facing clarity matters

7. AI System Strategy and Workflows | استراتيجية الذكاء الاصطناعي وتدفقاته

Recommended AI Positioning

The correct strategy is not to use AI everywhere. The correct strategy is to use AI where it adds clear educational or operational value and use normal logic where deterministic logic is enough.

القرار الموصى به: Gemini-first but provider-flexible. This means Gemini can be the initial provider, but the platform should expose an internal AI service layer so the provider can be changed later without rewriting the product.

MVP AI Roles

- Question generation for teachers as draft content, not directly published content
- Answer evaluation for suitable short-answer or open-ended items
- Feedback, explanation, and hint generation in controlled structured output
- Bilingual support through per-request language control

MVP AI Non-Goals

- No unrestricted chatbot tutor in the first MVP
- No voice tutoring or highly advanced adaptive engine initially

- No blind autonomous grading without backend validation and teacher control

AI Workflow Summary

Teacher-side AI flow	Student-side AI flow
Teacher submits generation request -> backend validates -> AI returns structured JSON -> backend validates -> questions saved as draft -> teacher reviews and approves	Student submits answers -> backend grades MCQ rule-based -> short answers sent to AI -> evaluation JSON returned -> backend validates -> score and feedback stored

AI Design Principles

- Backend-controlled orchestration; AI never writes directly to production logic unvalidated
- Structured JSON output only for machine-safe handling
- Teacher review before publish for generated content
- Separate logs for generation and evaluation events
- Cost-aware usage: do not spend AI calls where rules are sufficient

8. Data Model and Database Direction | اتجاه قاعدة البيانات ونموذج البيانات

Recommended Database Engine

PostgreSQL is the recommended database engine because it fits a relational educational SaaS product well, supports constraints cleanly, and scales better for structured queries and future analytics.

Core Entity Set

- users
- student_profiles
- teacher_profiles
- subjects
- topics
- questions
- question_options
- exams
- exam_questions
- student_exam_attempts
- student_answers
- answer_evaluations
- comments (optional in MVP if needed)
- ai_generation_logs
- ai_evaluation_logs

Key Database Principles

- Users + profile tables rather than unrelated duplicated identity tables
- Many-to-many relationship between exams and questions through a linking table
- Answers separated from evaluations to support AI, rules, and manual override
- Question bank structure from day one instead of one-off embedded exam text

Condensed ERD View

Entity Group	Relationship Intent
users -> student_profiles / teacher_profiles	Shared identity with role-specific data
subjects -> topics	Academic categorization for content and analytics
teacher_profiles -> exams	Teachers create and manage exams
exams <-> questions via exam_questions	Reusable question bank model
student_profiles -> student_exam_attempts -> student_answers	Attempt and answer tracking
student_answers -> answer_evaluations	Separate evaluation layer
AI logs	Track generation and evaluation events for debugging, quality, and cost control

9. Core User Flows | تدفقات المستخدم الرئيسية

Student Flow

Student logs in -> opens dashboard -> selects an exam -> starts attempt -> answers questions -> submits -> system grades / evaluates -> student reviews result and feedback

Teacher AI Generation Flow

Teacher selects topic, difficulty, question type, count, and language -> backend validates -> AI returns structured content -> backend validates -> questions saved as draft -> teacher edits/approves -> content used in exam bank

Manual Exam Creation Flow

Teacher creates exam metadata -> selects existing questions or approved AI drafts -> orders questions -> sets duration/status -> saves draft -> publishes

Admin Flow

Admin signs in -> reviews users and roles -> manages activation/deactivation and later broader system oversight

10. UX / UI Requirements | متطلبات واجهة وتجربة المستخدم

- Modern educational SaaS feel with strong clarity and restrained professional styling
- Responsive design from the beginning
- Arabic and English readiness, including RTL and LTR support
- Simple exam-taking flow with minimal confusion
- Teacher workflows should feel efficient and not overloaded
- Feedback should be presented as understandable learning guidance, not raw system noise

Primary Screens

- Landing / home page
- Login and register
- Student dashboard
- Teacher dashboard

- Exam page
- Results page
- Admin area
- Question-bank and exam-builder views

11. Non-Functional Requirements | المتطلبات غير الوظيفية

Category	Requirement	Why it matters
Performance	Fast enough loading and responsive core flows	User trust and exam usability
Security	Password hashing, role-based access, validated APIs	Protect users and system integrity
Scalability	Modular AI layer and normalized data model	Avoid costly rebuild later
Reliability	Safe attempt storage and recoverable AI error handling	Educational records cannot be fragile
Maintainability	Clear structure, naming, modular logic	Beginner-to-mid team needs discipline
Usability	Clean, bilingual-ready, low-friction interaction	Adoption and effectiveness

12. Team System and Execution Model | نظام الفريق وآلية التنفيذ

The project requires an execution system, not only a task list. Because the team is currently between beginner and intermediate levels, the correct model must combine learning, delivery, transparency, and accountability.

Recommended Structure

Track / Role	Main Responsibility	Suggested Notes
Project Lead	Vision, coordination, priorities, client alignment	Likely the user
Frontend / UI-UX Track	Pages, responsiveness, dashboards, bilingual layout	2 to 3 people
Backend / DB / AI Track	Auth, APIs, data model, AI integration, business logic	Around 3 people

Recommended Cadence

- Weekly task planning with clear owners and definition of done
- Short daily standup for visibility and blocker discovery
- Weekly review/demo where each member shows real output
- Proof-based reporting: code, screenshot, demo, schema, API response, or design evidence

Healthy Accountability Model

- Public task board visible to all
- Each task must have a clear owner

- No invisible work counts as finished work
- Repeated delay should change responsibility level and increase follow-up, not just produce vague excuses
- Recognition for strong delivery should be visible, not hidden

13. Roadmap and Build Order | خارطة الطريق وترتيب البناء

Phase	Build Focus	Expected Result
Phase 1	Auth, users, roles, base UI skeleton, DB foundation	Stable starting platform structure
Phase 2	Questions, exams, attempts, results basics	Core educational workflow operational
Phase 3	AI generation flow, AI evaluation flow, teacher review workflow	Differentiated MVP value appears
Phase 4	Dashboard polish, testing, cleanup, deployment preparation	Professional MVP readiness

14. Cost Structure | هيكل التكلفة

The project cost model should always be split into three distinct categories so decisions remain clear and financially disciplined.

Cost Type	Description	Examples
Operational / Running Costs	Costs paid after delivery to keep the platform running	Hosting, DB hosting, domain, AI API usage, backups
Development Costs	Costs incurred while building the product	AI tool subscriptions, design tools, paid services, collaboration tools
Team / Service Fee	What the builders charge for the work itself	Analysis, design, implementation, setup, support packages

15. Risks, Constraints, and Mitigation | المخاطر والقيود وكيفية الحد منها

Risk Area	Example Risk	Impact	Mitigation
AI	Malformed output or inconsistent grading	Breaks trust and workflow	Structured JSON, backend validation, teacher review, retries
Team	Skill gaps and uneven execution speed	Delay and fragile implementation	Weekly + daily system, pairing, proof-based reporting
Scope	Trying to build too much too early	MVP failure and confusion	Strong scope boundaries and phased build order

Finance	AI usage cost grows unexpectedly	Budget pressure	Use AI only where needed, logging, prompt discipline
Architecture	One-off shortcuts in data or code	Expensive rewrite later	Question bank model, modular AI layer, normalized DB

16. Open Decisions and Next-Step Recommendations | القرارات المفتوحة والتوصيات التالية

Open Decisions

- Final official brand name and branding language direction
- Final backend framework choice (for example Node.js/Express vs Django)
- Exact hosting and deployment plan for MVP
- Whether numerical-answer grading enters the first MVP or a later phase
- Whether comments/notes module is MVP or post-MVP
- Exact payment model and timing of gateway integration

Recommended Next Deliverables

- Turn this PRD into a full formal SRS or proposal version for client-facing approval
- Create a technical architecture diagram and API map
- Produce the database schema / ERD as a separate implementation document
- Prepare a cost model for MVP and Full Platform with Oman-oriented operational assumptions
- Create detailed track-by-track weekly execution plan for the team

Final Conclusion | الخلاصة النهائية

AI-Masarrat has strong potential if executed with discipline. The platform's advantage is not just the presence of AI, but the structured and educationally meaningful way AI is integrated into the system.

قوة المشروع ليست في استخدام الذكاء الاصطناعي فقط، بل في استخدامه داخل بنية تعليمية منضبطة: أسئلة أفضل، تقييم أوضح، متابعة أقوى، وتجربة أكثر فائدة للطلاب والمدرس.

The immediate objective should be a real, disciplined MVP: technically clean, educationally useful, financially aware, and operationally realistic.